

HARSH KUMAR

Backend & Full-Stack Developer

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PROFILE

Backend-focused Computer Science student building production-oriented applications with **microservices**, **event-driven architectures**, **APIs**, **databases**, **caching**, **real-time systems**, and **asynchronous processing**. Interested in scalable backend infrastructure, distributed systems, observability, and AI-powered services.

EDUCATION

Roorkee Institute of Technology, Roorkee

2023 – 2027

Bachelor of Technology (B.Tech), Computer Science & Engineering

Girish Prasad Memorial College

Class XII – 86.80% (2022) | Class X – 85.67% (2020)

TECHNICAL SKILLS

- **Languages:** C++, Python, TypeScript, JavaScript, SQL
- **Backend:** Node.js, Express.js, NestJS, FastAPI, tRPC, REST APIs, Microservices
- **Databases:** PostgreSQL, MongoDB, Redis, ClickHouse, Prisma, Drizzle ORM
- **Distributed Systems:** Apache Kafka, Redpanda, NATS JetStream, QStash, WebSockets, SSE
- **Frontend:** React.js, Next.js, Redux Toolkit, TanStack Query, Tailwind CSS
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

PROJECTS

Multi-Vendor E-Commerce Platform

GitHub

Tech: Next.js, TypeScript, Node.js, Express.js, MongoDB, Prisma, Redis, Kafka, Redpanda, Socket.IO, Nx

- Architected an Nx-based microservice platform supporting **3 roles**: buyers, sellers, and administrators.
- Engineered **Kafka + Redpanda** event pipelines for asynchronous order processing and activity tracking.
- Added **Redis caching + Socket.IO** for low-latency data access and real-time order updates.
- Implemented JWT/RBAC security, admin analytics, **TensorFlow recommendations**, and ImageKit media optimization.

ULogs – Observability Platform

GitHub

Tech: Next.js, TypeScript, NestJS, PostgreSQL, Drizzle, Redis, NATS JetStream, ClickHouse, Clerk, Stripe

- Built an observability platform across **3 layers**: dashboard, NestJS API, and typed SDK.
- Engineered **NATS JetStream + ClickHouse** pipelines for asynchronous log ingestion and analytics.
- Implemented **2 auth paths**, Redis quotas, Argon2 API keys, SSE streaming, and HMAC webhooks.
- Integrated Stripe billing and a typed SDK with buffered delivery, log reads, live streams, and graceful flushing.

MyTube – YouTube-Style Video Platform

GitHub

Tech: Next.js, TypeScript, React, tRPC, PostgreSQL, Drizzle, Mux, Upstash Redis, QStash, Clerk, Gemini

- Built a full-stack video platform covering uploads, streaming, channels, playlists, search, comments, and creator tools.
- Engineered **Mux + QStash** workflows for asynchronous video processing and durable background jobs.
- Implemented **3 Gemini workflows** for transcript-based title, description, and thumbnail generation.
- Developed **12 tRPC routers** with Clerk authentication and Redis sliding-window rate limiting.

ACHIEVEMENTS

- Solved **125+ LeetCode** and **605+ GeeksforGeeks** DSA problems.
- **1st Place – RIT-A-THON:** Won a 30-hour intra-level hackathon as part of a five-member team.